Online Test 1

(1) $3{,}78 \times 10^{-3} = L \times B \times H$

$3{,}78 \times 10^{-3} = 34 \times H$

$H = 1{,}111 \times 10^{-4}$

$0000001{,}111$ m

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(2) $\cancel{\Delta x = v}$

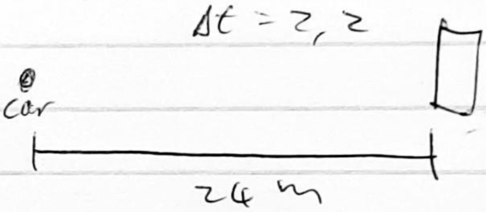
56 km/h $= 15{,}556$ m/s

$\Delta x = V_i \Delta t + \frac{1}{2} a (\Delta t)^2$

$24 = 15{,}556(2{,}2) + \frac{1}{2} a (2{,}2)^2$

$a = -4{,}22$ m/s^2

$\Delta t = 2{,}2$



$\cancel{\Delta x = \left(\frac{V_f + V_i}{2}\right) \Delta t}$

$=$

$V_f^2 = V_i^2 + 2a\,\Delta x$

$= (15{,}556)^2 + 2(-4{,}22)(24)$

$V_f^2 = 6{,}28$ m/s

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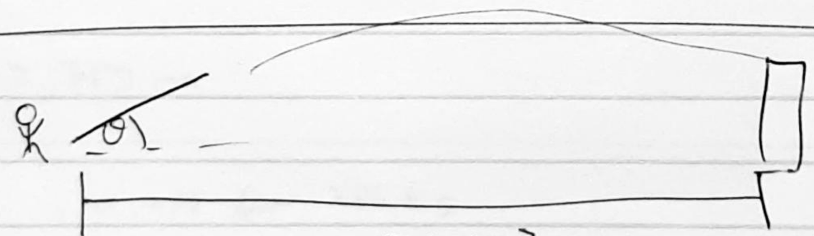
(3)



$\Delta x = 324{,}2$ m

$t_f = 5{,}64$ s

$t_i = 0$

$V_i = 70{,}6$ m/s

$\theta = 36{,}776^\circ$

a) $\cancel{V_x = V_i \cos\theta}$

$0{,}953 = \sin^2\theta$

$\sin\theta = 0{,}976$

$\theta = 77{,}422^\circ$

$h = \frac{V_i^2 \sin^2\theta}{2g}$

$242{,}317 = \frac{(70{,}6)^2 \sin^2\theta}{19{,}6}$

$\Delta y = V_i \Delta t + \frac{1}{2} g (\Delta t)^2$

$= (70{,}6)(5{,}64) + \frac{1}{2}(-9{,}8)(5{,}64)^2$

$= 242{,}317$ m

a) $V_x = V_i \cos\theta$

$= (70,6)\cos(36,776)$

$= 56,5$ m/s

b) $V_y = V_i \sin\theta$

$= (70,6)\sin(36,776)$

$= 42,3$ m/s

c) 36,8°

e) $h = \dfrac{v_i^2 \sin^2\theta}{2g}$

$= \dfrac{(60,6)^2 \sin^2(36,776)}{2(9,8}$

$= 91,3$ m

d) $\Delta y = V_i \Delta t + \frac{1}{2}(-9,8)(\Delta t)^2$

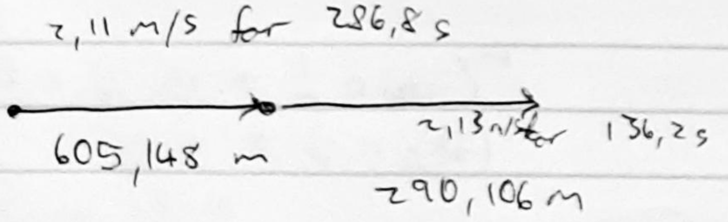
$91,1 = (70,6)\Delta t + \frac{1}{2}(-9,8)(\Delta t)^2$

$0 = -4,9\Delta t^2 + 70,6\Delta t - 91,3$

$\therefore t = 1,44$ s   or $t = 12,9$

f.) 242,317 m

(4)



$V = \dfrac{895,254}{423}$

$= 2,12$ m/s

~~4.~~

$V_i = 0$
$V_f = 21,5$
$a = 0,5$
$\Delta t = ?$

5 a) ~~21,5~~ $V_f = V_i + a\Delta t$

$21,5 = 0 + (0,5)\Delta t$

$21,5 = (0,5)\Delta t$

$t = 43s$

police — car

$$v_i \Delta t + \frac{1}{2}a(\Delta t)^2 = v_i\Delta t + \frac{1}{2}a(\Delta t)^2$$

$$0 + \frac{1}{2}(0,5)\Delta t^2 = (21,5)\Delta t + 0$$

$$\frac{1}{4}\Delta t^2 = 21,5\,\Delta t$$

$$\frac{\frac{1}{4}\Delta t^2}{\Delta t} = 21,5$$

$$\frac{1}{4}\Delta t = 21,5$$

$$=$$

$$\frac{1}{4}\Delta t^2 = 21,5\,\Delta t$$

$$\frac{1}{4}\Delta t^2 - 21,5\Delta t = 0$$

$$\Delta t\left(\frac{1}{4}\Delta t - 21,5\right) = 0$$

$$\frac{1}{4}\Delta t - 21,5 = 0$$

$$\frac{1}{4}\Delta t = 21,5$$

$$\Delta t = 86\,s$$

b) $V_f = V_i + a\Delta t$

$= 0 + (0,5)(86)$

$= 43\ m/s$

c) $\Delta x = V_i\,\Delta t + \frac{1}{2}a(\Delta t)^2$

$= 0 + \frac{1}{2}(0,5)(86)^2$

$= 1849$

8

$v_i = 0$

10

$$F = \frac{mv^2}{R}$$

$$= \frac{(525)(17)^2}{25} + (525)(9{,}8)$$

$$= 11214$$

A

$m_{TOT} = \cancel{571}\ 525\ kg$

$v = 17\ m/s$

$R = 25\ m$

$$509{,}6 = \frac{(525)(17)^2}{R} + (525)(9{,}8)$$

$$R = 32{,}73\ m$$